

Financial Analysts Journal

ISSN: 0015-198X (Print) 1938-3312 (Online) Journal homepage: www.tandfonline.com/journals/ufaj20

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To cite this article: Aleksandar Andonov, Florian Bardong & Thorsten Lehnert (2010) TIPS, Inflation Expectations, and the Financial Crisis, Financial Analysts Journal, 66:6, 27-39, DOI: 10.2469/faj.v66.n6.1

To link to this article: https://doi.org/10.2469/faj.v66.n6.1

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Published online: 31 Dec 2018.

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Financial Analysts Journal
Volume 66 • Number 6
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FAJ

TIPS, Inflation Expectations, and the Financial Crisis

Aleksandar Andonov, Florian Bardong, and Thorsten Lehnert

The authors show that inefficiencies in the U.S. market for inflation-linked bonds can be exploited by informed traders who include survey estimates or inflation model forecasts in trades on breakeven inflation. The Treasury Inflation-Protected Securities market has yet to fulfill investors' expectations as a low-risk, efficient, and liquid financial instrument.

Index-linked government bonds eliminate not only default risk but also inflation risk because they adjust cash flows for accrued inflation over time. In the past, the governments of many developed countries (e.g., the United Kingdom, Canada, and Sweden) issued inflation-linked bonds. In 1997, the U.S. Department of the Treasury followed suit by issuing Treasury Inflation-Protected Securities (TIPS). Since then, market liquidity for index-linked government bonds has improved markedly because these securities constitute an increasing share of government debt. As of June 2009, the market had grown to more than $532 billion, or 8 percent of marketable debt, corresponding to 4.6 percent of the total Treasury debt outstanding.[1]

We investigated the TIPS market to improve our understanding of the major factors that drive risk and returns in that asset class. Although earlier studies identified liquidity premiums as one possible if not major factor driving TIPS returns (see, e.g., Bardong and Lehnert 2004a), the financial crisis of 2008 allowed us to examine how risk premiums related to index-linked government bonds are affected by changes in the perception and pricing of risk. Moreover, unlike previous empirical studies of the TIPS market, the time horizon of our study included periods of macroeconomic volatility and market uncertainty. For example, Roll (2004) documented a significant decrease in TIPS yields toward the end of 2003, suggesting a change in the TIPS market environment. Not much later, the third quarter of 2008 saw the highest U.S. inflation rate since TIPS were first issued. These additional observations allowed us to explore how macroeconomic uncertainty and changes in inflation expectations affect investors in index-linked bonds and whether the inefficiencies documented in earlier studies (e.g., Barr and Campbell 1996) still exist. Thus, we examined the empirical question whether TIPS reduce or increase risks for the average investor.

Building on previous work by Bardong and Lehnert (2004a, 2004b, 2008)—who looked at excess returns of index-linked government bonds over comparable nominal bonds to assess the efficiency and risk characteristics of the former asset class—we compared breakeven inflation, a measure of inflation expectations commonly used in the TIPS market, with different inflation forecasts over a variety of horizons. We then looked at multiple ways to incorporate inflation expectations into TIPS investments, which allowed us to identify potential risk and return factors that TIPS investors face.

Literature Review

Conventional bonds provide nominal fixed-interest payments and a set principal at maturity. The real value of these cash flows is unknown when a nominal bond is issued because of uncertainty about future inflation. Therefore, inflation risk—the risk of unanticipated changes in the purchasing power of cash flows from nominal bonds—affects nominal bond issuers and investors (Wrase 1997). If inflation turns out to be higher than expected, nominal bond investors lose because the real value of their bond investment declines. If, however, inflation turns out to be lower than expected, bond issuers lose because the real cost of paying off their debt is higher than expected.

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Note: The views and opinions expressed in this article are the authors' own and do not necessarily reflect the views and opinions of BlackRock.

With an inflation-indexed bond, the real interest rate is set in advance because the nominal rate of return varies with the inflation rate over the life of the bond. Therefore, neither issuers nor investors face inflation risk (Wrase 1997), which makes inflation-indexed bonds attractive to investors who desire predictable real cash flows. At the same time, issuers of inflation-indexed bonds are also meant to benefit because the inflation indexation removes a potential inflation risk premium that would otherwise be added to the nominal interest payments of the bond.

In practice, however, benefits to issuers have been rather muted. Yields on TIPS have been unexpectedly high compared with those on comparable nominal Treasury bonds, resulting in larger-than-expected borrowing costs. The expected rate of inflation implied by TIPS has been about 50 bps below long-run inflation expectations (Sack and Elsasser 2004), which suggests that the interest costs of the issuer have been consistently higher than necessary to compensate for inflation. Carlstrom and Fuerst (2004) also found that the expected inflation rate derived from TIPS underestimates the actual inflation rate. Dudley, Roush, and Steinberg Ezer (2009), however, argued that the potential of TIPS to reduce borrowing costs depends on whether one analyzes the issue *ex ante* or *ex post*. Although the *ex post* results are not encouraging, Dudley et al. found that on an *ex ante* basis, the costs of issuing TIPS are equivalent to, if not lower than, the costs of issuing nominal Treasuries.

Sack and Elsasser (2004) offered three explanations for the low pricing of TIPS relative to their nominal equivalents. First, the low valuation of TIPS could reflect the small size of the TIPS investor community during the early years of that market. Second, supply-and-demand trends affected the relative values of nominal and real bonds. Although the supply of TIPS increased steadily over time, that was not the case for nominal Treasuries during the late 1990s. Treasuries thereby became relatively scarce compared with TIPS, which may have put upward pressure on TIPS yields. Third, the lower liquidity of TIPS relative to nominal Treasuries may have led investors to demand a liquidity premium for holding TIPS. This premium could be expected to decline over time because liquidity has improved in recent years. In all probability, however, TIPS will not achieve the same liquidity as nominal Treasuries because of the different roles they play in financial markets (Sack and Elsasser 2004). TIPS are primarily used by long-term buy-and-hold investors, whereas nominal securities are largely used for hedging and active trading.

With respect to the liquidity of inflation-indexed government bonds, the U.S. experience is in line with that of other countries. Reflecting the importance of buy-and-hold investors, index-linked bond markets in other countries also tend to be relatively small, with low trading activity (Wrase 1997). As a result, index-linked bond prices are likely to be driven by factors other than expected inflation, such as liquidity premiums.

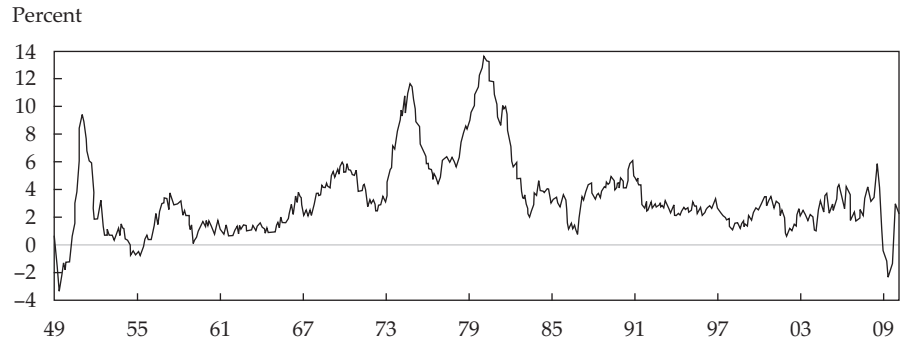
In fact, TIPS prices may provide a biased estimate of inflation expectations. Carlstrom and Fuerst (2004) found that in the absence of liquidity risk, the expected inflation implied in TIPS overstates actual inflation expectations by about 95 bps. This finding suggests that inflation risk premiums are incorporated in nominal bond prices, which Campbell and Shiller (1996) estimated to be between 50 and 100 bps. Further, Sack (2000) found considerable variation in inflation risk premiums. Therefore, TIPS prices appear to be driven by such factors as inflation expectations, liquidity, and inflation risk premiums.

Bardong and Lehnert (2004a, 2004b, 2008) went one step further by focusing on not only the price process of inflation-linked bonds but also the potential for using signals to indicate future returns. The signals they used were based on the inflation-forecasting models of McCulloch and Stec (2000) and Kothari and Shanken (2004) and inflation forecasts by the Survey of Professional Forecasters (Survey). Because the data that underlie the inflation models and the Survey are publicly available, one would not expect systematic excess returns from using such data as trading signals if the TIPS market is "semi-strong efficient."[2] The findings by Bardong and Lehnert (2004a, 2004b, 2008) suggest that investors are able to earn, on average, positive excess returns, which is an argument against TIPS market efficiency.

Bardong and Lehnert (2004a, 2004b, 2008) studied the six years after TIPS were first issued, in 1997. Given the lack of liquidity in the early years of the TIPS market, their results were conceivably driven by liquidity premiums. Liquidity and investors' understanding of the TIPS market, however, have improved in recent years. Therefore, the profit opportunities that Bardong and Lehnert documented could have disappeared. At the same time, the TIPS market experienced a great deal of volatility. For instance, Roll (2004) found that TIPS yields decreased significantly between 2000 and 2003, whereas inflation volatility has increased in recent years (**Figure 1**).

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Figure 1. Annual Inflation Rate, January 1949–December 2009



Notes: This figure presents the year-on-year inflation rate calculated on the basis of the monthly values of the CPI for All Urban Consumers (CPI-U) for 1997–2009. We retrieved the monthly values of the CPI-U from the U.S. Department of Labor, Bureau of Labor Statistics.

From July 2008 to May 2009, U.S. inflation went from a 17-year high of more than 5 percent to a 40-year low of below –2 percent. This macroeconomic volatility was occurring at the same time that the financial crisis was causing a rapid repricing of risk premiums. Both the volatility and the repricing of risk premiums strongly affected the TIPS market environment, which may have led to profit opportunities for TIPS investors who understood the main drivers of risk and return in that particular market.

According to economic theory and experience, investors move their funds into safe and liquid assets during crises. Although TIPS provide the safety of a government-backed instrument, they are less liquid than nominal government bonds, as previous research has shown (see, e.g., Sack and Elsasser 2004). We looked at the impact of the recent financial crisis on the behavior of the TIPS market and the possibility of implementing profitable trading strategies. Thus, recent improvements in TIPS liquidity and market depth, together with the unprecedented volatility of real yields and inflation, provided the setting for our extension of previous empirical research.

Research Design

Our research design encompassed possible trading strategies and considered a calculation of breakeven inflation, together with data from the Survey of Professional Forecasters and an adjusted inflation-forecasting model.

Trading Strategies. The inflation adjustment to an index-linked bond is the factor that most directly affects its cash flows. If the market-implied inflation rate differs from the realized rate, the price of an inflation-indexed bond is adjusted accordingly because its cash flow is directly related to the realized rate of inflation. Therefore, if an investor estimates the future inflation rate more precisely than the market does, she should earn consistent excess returns. Consistent excess returns, however, imply market inefficiencies. Therefore, TIPS market efficiency can be tested by using the difference between the market-implied inflation rate and a set of forecasts as trading signals.

We tested these signals by using a variety of trading strategies. Assuming that our alternative sources of information were more accurate in forecasting inflation than were market expectations, we interpreted a forecasted inflation that was higher than market-implied inflation as a signal to buy (overweight) TIPS. Conversely, if forecasted inflation was lower than market-implied inflation, we shorted (underweighted) TIPS.

In addition to holding TIPS outright, we also looked at trading TIPS in a way that was specific to the inflation-linked bond market. This so-called breakeven trading strategy aims to capture changes in the market-implied rate of inflation over time. By isolating the source of returns to changes in the market-implied rate of inflation, one can ensure that other return factors that affect TIPS prices, such as the real yield, are neutralized (Bardong and Lehnert 2004a). A long breakeven position is implemented by going long on an index-linked bond and by going short on a maturity-matched nominal bond. As with the long–short TIPS strategy, we took a long breakeven position if the forecasted inflation exceeded market-implied inflation.

The alternative holding periods for each trading strategy were 3, 6, and 12 months. At the end of each holding period, we cleared the positions and compared the returns with an adequate benchmark. We compared the returns on the long–short

strategy with the returns on a buy-and-hold strategy over the same holding period. Because the breakeven trading strategy had a different risk profile, we compared it with a long breakeven position.

We tested the profitability of the trading strategies both with and without trading costs. We assumed trading costs of 11 bps (see Bardong and Lehnert 2004a).

Calculation of Breakeven Inflation. By market convention, the difference between the nominal yield on a Treasury bond and the real yield on a maturity-matched index-linked bond is referred to as breakeven inflation (BEI). BEI is used to measure the inflation expectations of financial markets.

BEI may not always be a useful predictor of future inflation; as a short-term inflation signal, it has several limitations. First, BEI is theoretically an estimate of the average inflation rate until the maturity of the respective TIPS, which does not take into account cyclical aspects of inflation. For instance, TIPS are based on the CPI for All Urban Consumers, which is not adjusted for seasonal changes in the price level. Second, nominal bond yields include compensation for expected inflation, as well as an inflation risk premium. If the inflation risk premium is significant, BEI will overstate inflation expectations. Despite these caveats, BEI is considered a reasonable estimate of the market consensus on future inflation.

BEI is calculated daily for each TIPS issue by using the Fisher equation. The Fisher equation states that the nominal yield comprises the real yield and inflation compensation:

$$1 + y^n = \left(1 + y^r\right)\left[1 + E(i)\right], \tag{1}$$

where y^n, y^r, and $E(i)$ represent the nominal yield, the real yield, and inflation compensation, respectively. Nominal interest rates are taken from maturity-matched nominal bonds.[3] Therefore, we matched all TIPS with STRIPS (Separate Trading of Registered Interest and Principal Securities) that had the closest maturity date.[4] **Table 1** presents the matching of all 32 series of TIPS and STRIPS. The typical maturity mismatch between TIPS and their closest maturity-matched nominal bonds is one month. We retrieved the daily price for each series from Thomson Reuters DataStream. The period for our study begins with the issuance of the first series of TIPS (30 January 1997) and ends in June 2009.[5]

One observation that can be made from the data is that the returns on TIPS are less volatile than the returns on the matching nominal bonds. Nominal yields are much more volatile than real yields because changes in nominal yields incorporate shocks to expected inflation (Roll 2004). Developing this point further, Kothari and Shanken (2004) found that the real returns on TIPS are less volatile than their nominal returns, which is the opposite of what one observes for nominal bonds. The lower return volatility of TIPS, however, applies only to those series that have not yet matured. The four TIPS series that have already matured show a higher standard deviation of returns. This finding can be explained by the fact that TIPS do not provide inflation protection for the last three months of their lives because of the inflation indexation lag.[6]

BEI can be calculated from the matched TIPS and STRIPS series. BEI consists of expected inflation and risk premiums related to inflation and potentially other factors. In practice, changes in the inflation risk premium are hard to distinguish from changes in expected inflation (Spiegel 1998). Hence, most researchers usually assume that the inflation risk premium is equal to zero (Arak and Kreicher 1985; Woodward 1990; Barr and Campbell 1996). This assumption is typically justified by the small size of the empirical estimates of the inflation risk premium (see, e.g., Cogley 1995).

Daily BEI can be calculated for each pair of bonds by using daily values of real and nominal yields. These daily BEI rates can then be compared with the forecasts from the Survey or with the forecasts from the inflation-forecasting model presented later in the article.

Survey of Professional Forecasters. The Survey was started in 1968 by the American Statistical Association and the National Bureau of Economic Research. Initially, the Survey numbered around 60 forecasters, which decreased to about 14 in the 1970s and 1980s. The Federal Reserve Bank of Philadelphia took over the management of the Survey in 1990. Since then, the number of forecasters has stabilized at around 30. One-year-ahead inflation expectations are annualized averages over four quarters, beginning with the quarter following the Survey date. The inflation forecasts are released at 10:00 a.m. local time, which implies that they are price relevant for that particular day (Bardong and Lehnert 2008). The inflation forecasts go back to the fourth quarter of 1994.[7]

The Survey's main strength is that the forecasters are close to economic decision makers, which suggests that the Survey reflects beliefs that affect important investment and pricing decisions. Forecasters may provide biased predictions owing to, for example, strategic considerations (Giordani and Söderlind 2003). Another weakness of the Survey is that it reflects the beliefs of a small fraction of market participants, which may not be representative

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Table 1. Matching of TIPS with Nominal Bonds of Comparable Maturity

	TIPS				Nominal Bonds			
Name	Start of Time Series	Maturity Date	Avg. Real Return (%)	Std. Dev. (%)	Name	Maturity Date	Avg. Nominal Return (%)	Std. Dev. (%)
TIPS 3.375	30 Jan 1997	15 Jan 2007	2.70	1.73	STRIPS	15 Feb 2007	4.62	1.41
TIPS 3.625	12 Jan 1998	15 Jan 2008	2.56	1.27	STRIPS	15 Feb 2008	4.56	1.16
TIPS 3.875	7 Jan 1999	15 Jan 2009	2.64	2.71	STRIPS	15 Feb 2009	4.29	1.42
TIPS 4.250	18 Jan 2000	15 Jan 2010	2.22	1.27	STRIPS	15 Feb 2010	4.05	1.45
TIPS 0.875	27 Oct 2004	15 Apr 2010	1.88	1.24	STRIPS	15 May 2010	3.39	1.47
TIPS 3.500	16 Jan 2001	15 Jan 2011	2.04	0.98	STRIPS	15 Feb 2011	3.98	1.20
TIPS 2.375	26 Apr 2006	15 Apr 2011	1.88	1.07	STRIPS	15 May 2011	3.07	1.49
TIPS 3.375	10 Jan 2002	15 Jan 2012	1.93	0.77	STRIPS	15 Feb 2012	4.00	1.03
TIPS 2.000	25 Apr 2007	15 Apr 2012	1.59	0.93	STRIPS	15 May 2012	2.74	1.09
TIPS 3.000	11 Jul 2002	15 Jul 2012	1.86	0.66	STRIPS	15 Aug 2012	3.97	0.93
TIPS 0.625	23 Apr 2008	15 Apr 2013	1.32	0.57	STRIPS	15 May 2013	2.47	0.71
TIPS 1.875	10 Jul 2003	15 Jul 2013	1.88	0.59	STRIPS	15 Aug 2013	4.05	0.93
TIPS 2.000	9 Jan 2004	15 Jan 2014	1.92	0.59	STRIPS	15 Feb 2014	4.10	0.87
TIPS 1.250	24 Apr 2009	15 Apr 2014	1.11	0.11	STRIPS	15 May 2014	2.67	0.31
TIPS 2.000	9 Jul 2004	15 Jul 2014	1.94	0.58	STRIPS	15 Aug 2014	4.11	0.84
TIPS 1.625	14 Jan 2005	15 Jan 2015	2.01	0.58	STRIPS	15 Feb 2015	4.17	0.79
TIPS 1.875	15 Jul 2005	15 Jul 2015	2.05	0.58	STRIPS	15 Aug 2015	4.18	0.79
TIPS 2.000	13 Jan 2006	15 Jan 2016	2.10	0.58	STRIPS	15 Feb 2016	4.23	0.75
TIPS 2.500	18 Jul 2006	15 Jul 2016	2.06	0.59	STRIPS	15 Aug 2016	4.19	0.66
TIPS 2.375	12 Jan 2007	15 Jan 2017	2.02	0.59	STRIPS	15 Feb 2017	4.15	0.62
TIPS 2.625	13 Jul 2007	15 Jul 2017	1.90	0.56	STRIPS	15 Aug 2017	4.05	0.56
TIPS 1.625	11 Jan 2008	15 Jan 2018	1.82	0.53	STRIPS	15 Feb 2018	3.90	0.47
TIPS 1.375	11 Jul 2008	15 Jul 2018	3.01	0.46	STRIPS	15 Aug 2018	3.84	0.49
TIPS 2.125	7 Jan 2009	15 Jan 2019	1.72	0.19	STRIPS	15 Feb 2019	3.63	0.33
TIPS 2.375	28 Jul 2004	15 Jan 2025	2.23	0.30	STRIPS	15 Feb 2025	4.84	0.43
TIPS 2.000	25 Jan 2006	15 Jan 2026	2.32	0.29	STRIPS	15 Feb 2026	4.80	0.47
TIPS 2.375	24 Jan 2007	15 Jan 2027	2.30	0.31	STRIPS	15 Feb 2027	4.69	0.48
TIPS 1.750	25 Jan 2008	15 Jan 2028	2.25	0.33	STRIPS	15 Feb 2028	4.47	0.47
TIPS 3.625	14 Apr 1998	15 Apr 2028	2.86	0.74	STRIPS	15 May 2028	5.06	0.51
TIPS 2.500	27 Jan 2009	15 Jan 2029	2.31	0.15	STRIPS	15 May 2029	4.32	0.32
TIPS 3.875	8 Apr 1999	15 Apr 2029	2.77	0.73	STRIPS	15 May 2029	5.17	0.53
TIPS 3.375	11 Oct 2001	15 Apr 2032	2.33	0.45	STRIPS	15 Feb 2031	5.03	0.52

Notes: This table shows TIPS and their nominal counterparts matched by maturity date, together with some summary statistics. Under the TIPS header, *Name* shows the semi-annual coupon paid for each TIPS issue and *Start of Time Series* refers to the date when the bond was issued. Under the Nominal Bonds header, the STRIPS series usually started before the TIPS and, therefore, only the maturity date is reported.

of the market as a whole. Because the Survey is publicly available, it should be reflected in TIPS prices if it is informative and if the TIPS market is semi-strong efficient.

The Survey can be used for up to one-year-ahead inflation forecasts. To generate longer-range forecasts, we used an inflation-forecasting model.

Inflation-Forecasting Model. Kothari and Shanken (2004) presented a model for forecasting inflation up to three years into the future. They explained current inflation with past financial and economic variables. The estimated coefficients were then used to forecast future inflation on the basis of the current values of the explanatory variables. For our study, we used the Kothari–Shanken inflation-forecasting model as adjusted for daily frequency by Bardong and Lehnert (2004a).

In this inflation-forecasting model, the explanatory variables are chosen on the basis of previous empirical research. The past one-year spot rate is used because interest rates have been found to capture inflation expectations (Fama 1975). Because real interest rates are likely to change over time, additional variables are needed to improve the explanatory power of the model. The yield spread

between five-year zero-coupon bonds and the one-year spot rate is considered a proxy for general business conditions (Fama and French 1989) and should thus be correlated with expected real interest rates. One-year-lagged inflation is included because Nelson and Schwert (1977) found that past inflation has incremental explanatory power beyond the spot interest rate for forecasting near-term inflation. Finally, the sum of real T-bill returns represents a direct if noisy proxy for realized real interest rates. Assuming that the real rate changes slowly over time, this variable could provide the model with additional explanatory power. Therefore, the following regression model is estimated:

$$Inf_t = \beta_0 + \beta_1 1YrSpot_{t-1} + \beta_2 5-1Spread_{t-1} + \beta_3 Inf_{t-1} + \beta_4 RR_{t-1} + \varepsilon_t, \quad (2)$$

where

Inf_t = inflation at time t

$1YrSpot$ = the nominal one-year spot interest rate

$5–1Spread$ = the yield spread between a five-year zero-coupon bond and the one-year spot rate

RR_{t-1} = the sum of realized monthly returns on one-month T-bills over the preceding one-year period observed in the previous period

The coefficients from this regression are then used to forecast inflation at time $t + 1$ with the right-hand variables at time t. One critical assumption is that the relationship between inflation and the explanatory variables changes only slowly over time.

The one-year spot rate and the five-year nominal yield on zero-coupon bonds (both from the constant maturity curve) are available from DataStream on a daily basis. Inflation is reported monthly and is thus converted into a daily frequency.[8]

The sum of the realized monthly returns on government securities over a one-year period, RR_{t-1}, is calculated as

$$RR_t = YTM_{t-12} + \sum_{i=1}^{12}\left[\left(\frac{1+YTM_i}{1+YTM_{i+1}}\right)^{1/12} - 1\right], \quad (3)$$

where YTM_{t-12} is the annualized yield to maturity observed in the previous year and YTM_i is the annualized yield to maturity observed at the beginning of month i. This expression is calculated by using daily values of the one-year constant maturity zero-yield curve from DataStream.

The regression results are presented in **Table 2**.

Table 2. Inflation Regressions, September 1990–July 2009

Variable	Coefficient	Std. Error	t-Statistic[a]
Intercept	0.038	0.0024	15.67
$1YrSpot_{t-1}$	0.264	0.0458	5.77
$5–1Spread_{t-1}$	0.099	0.0800	1.24
Inf_{t-1}	–0.436	0.0693	–6.30
RR_{t-1}	–0.268	0.0305	–8.77
Adjusted R^2 = 23.0%			

Note: This table shows summary statistics of the Kothari–Shanken (2004) inflation model as specified in Equation 2.

[a]Reflects a Newey–West adjustment with nine lags.

As theory predicts, the one-year spot interest rate is positively related to next year's inflation. The real return is negatively related to one-year-ahead inflation. This finding can be explained by the fact that—keeping the spot rate constant—a higher expected real interest rate implies a lower expected inflation rate. The negative coefficient of lagged inflation suggests mean reversion, which is in line with previous findings (Kothari and Shanken 2004). Overall, our explanatory variables explain around 23 percent of the variation in next year's inflation between 1990 and 2009, which is comparable with the findings of Kothari and Shanken (2004), who looked at 1953–2000.

Results

We analyzed the profitability of our trading strategies by using five inflation-forecasting horizons: the current quarter (Q_0), the next quarter (Q_1), the next year (Y_1), two years ahead (Y_2), and three years ahead (Y_3). In addition, we considered three different holding periods: 3 months, 6 months, and 12 months. Finally, we looked at each combination of forecast horizon and holding period both with and without transaction costs.

Table 3 presents the results for the long–short strategy with inflation forecasts from the Survey for the current quarter.

The long–short strategy shows positive excess returns on the first series of TIPS for the three-month holding period. This trading strategy was profitable for all TIPS series that had already matured. As one moves closer to the series with longer maturities and less data history, the negative performance of the long–short strategy becomes more pronounced. The same conclusions can be drawn for the holding periods of 6 and 12 months. Although the performance across the different holding periods exhibits little variation, the losses from the long–short strategy are lowest for the six-month holding period.

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Table 3. Performance of Long–Short Strategy with Q_0 Forecasting Horizon

		ER with 3-Mo. HP		ER with 6-Mo. HP		ER with 12-Mo. HP	
Name	Maturity Date	Without TC (%)	With TC (%)	Without TC (%)	With TC (%)	Without TC (%)	With TC (%)
TIPS 3.375	15 Jan 2007	2.02	1.60	2.01	1.58	3.08	2.65
TIPS 3.625	15 Jan 2008	1.51	1.00	1.22	0.70	1.89	1.38
TIPS 3.875	15 Jan 2009	2.23	1.54	2.52	1.82	2.33	1.62
TIPS 4.250	15 Jan 2010	1.42	0.91	1.50	0.99	1.46	0.95
TIPS 0.875	15 Apr 2010	2.30	1.71	2.63	2.00	2.39	1.71
TIPS 3.500	15 Jan 2011	1.09	0.58	1.25	0.73	1.75	1.23
TIPS 2.375	15 Apr 2011	–0.68	–1.12	0.38	–0.13	0.38	–0.13
TIPS 3.375	15 Jan 2012	–0.18	–0.79	0.07	–0.52	–0.19	–0.82
TIPS 2.000	15 Apr 2012	0.00	0.00	0.00	0.00	0.00	0.00
TIPS 3.000	15 Jul 2012	1.30	0.60	–0.10	–0.81	1.64	0.93
TIPS 0.625	15 Apr 2013	–1.26	–2.19	–1.59	–2.51	–4.60	–5.52
TIPS 1.875	15 Jul 2013	–0.42	–0.96	–0.05	–0.59	0.01	–0.52
TIPS 2.000	15 Jan 2014	–0.80	–1.19	0.02	–0.36	–0.45	–0.83
TIPS 1.250	15 Apr 2014	9.57	7.31	9.57	7.31	9.57	7.31
TIPS 2.000	15 Jul 2014	–1.68	–2.28	–1.27	–1.89	–1.19	–1.84
TIPS 1.625	15 Jan 2015	–0.83	–1.69	0.11	–0.76	0.60	–0.32
TIPS 1.875	15 Jul 2015	–0.28	–1.28	0.55	–0.48	0.55	–0.51
TIPS 2.000	15 Jan 2016	–0.64	–1.79	–0.06	–1.24	0.22	–1.02
TIPS 2.500	15 Jul 2016	–5.56	–6.68	–4.42	–5.54	–5.00	–6.12
TIPS 2.375	15 Jan 2017	–5.39	–6.61	–4.60	–5.82	–4.63	–5.84
TIPS 2.625	15 Jul 2017	–5.93	–6.95	–4.97	–5.99	–4.97	–5.99
TIPS 1.625	15 Jan 2018	4.35	3.44	5.35	4.44	5.35	4.44
TIPS 1.375	15 Jul 2018	0.00	0.00	0.00	0.00	0.00	0.00
TIPS 2.125	15 Jan 2019	–0.75	–1.49	–3.25	–3.99	–3.25	–3.99
TIPS 2.375	15 Jan 2025	–6.36	–6.71	–5.73	–6.09	–5.69	–6.05
TIPS 2.000	15 Jan 2026	–5.27	–5.84	–5.19	–5.76	–5.12	–5.64
TIPS 2.375	15 Jan 2027	–7.07	–7.39	–7.14	–7.46	–7.14	–7.46
TIPS 1.750	15 Jan 2028	3.01	2.10	3.62	2.71	3.57	2.65
TIPS 3.625	15 Apr 2028	–3.84	–4.16	–4.25	–4.77	–5.39	–5.94
TIPS 2.500	15 Jan 2029	–19.66	–20.90	–18.07	–19.30	–18.07	–19.30
TIPS 3.875	15 Apr 2029	–3.70	–3.99	–4.09	–4.49	–4.51	–4.92
TIPS 3.375	15 Apr 2032	–4.31	–4.65	–4.40	–4.72	–5.66	–6.01

Notes: This table presents the excess returns (ER) of the long–short trading strategy on the basis of the Q_0 forecasting horizon and different holding periods. The column *Name* provides the coupon of each TIPS issue. Excess returns for each holding period (HP) are presented with and without transaction costs (TC).

Using the Q_1 forecasting horizon does not dramatically change the results (not reported). Again, the long–short strategy is profitable only for the first series of TIPS, and negative returns persist across the other series. Therefore, Survey inflation forecasts for Q_0 and Q_1 do not seem to be informative for a long–short trading strategy.

Table 4 presents the results for the long–short strategy under the Kothari–Shanken (2004) model.

The long–short strategy performs better under this inflation-forecasting model. For instance, with the one-year-ahead inflation forecast, positive excess returns are achieved across all three holding periods. The strongest performance is shown for the shortest holding period (three months). Once trading costs are accounted for, however, positive excess returns disappear. Results for two- and three-years-ahead inflation forecasts are similar.

The first important observation from comparing the profitability of Survey forecasts with the profitability of our forecasting model is that the long–short strategy is profitable only for TIPS of very short maturity and for TIPS that have already matured. In addition, signals based on the forecasting model in conjunction with the long–short trading strategy show excess returns for the intermediate section of the TIPS maturity spectrum. Regardless of the source of the inflation forecasts, however, the long–short trading strategy leads to negative returns when it is used with longer-maturity TIPS.

Table 4. Performance of Long–Short Strategy with Y_1 Forecasting Horizon

		ER with 3-Mo. HP		ER with 6-Mo. HP		ER with 12-Mo. HP	
Name	Maturity Date	Without TC (%)	With TC (%)	Without TC (%)	With TC (%)	Without TC (%)	With TC (%)
TIPS 3.375	15 Jan 2007	0.66	0.24	0.46	0.03	2.12	1.69
TIPS 3.625	15 Jan 2008	0.23	–0.21	0.17	–0.27	0.59	0.15
TIPS 3.875	15 Jan 2009	0.79	0.39	0.78	0.39	0.78	0.39
TIPS 4.250	15 Jan 2010	–0.38	–0.81	–0.96	–1.40	–1.08	–1.53
TIPS 0.875	15 Apr 2010	0.08	–0.15	0.00	–0.23	0.10	–0.13
TIPS 3.500	15 Jan 2011	–1.15	–1.89	–1.67	–2.41	–3.45	–4.19
TIPS 2.375	15 Apr 2011	0.00	0.00	0.00	0.00	0.00	0.00
TIPS 3.375	15 Jan 2012	–3.61	–4.25	–3.60	–4.24	–3.77	–4.40
TIPS 2.000	15 Apr 2012	–0.38	–0.49	–0.38	–0.49	–0.38	–0.49
TIPS 3.000	15 Jul 2012	–1.72	–2.29	0.52	–0.05	0.52	–0.05
TIPS 0.625	15 Apr 2013	–0.10	–0.75	–0.43	–1.07	–2.74	–3.38
TIPS 1.875	15 Jul 2013	1.25	0.72	1.28	0.75	1.28	0.75
TIPS 2.000	15 Jan 2014	0.76	0.11	0.79	0.14	0.51	–0.13
TIPS 1.250	15 Apr 2014	13.45	11.76	13.45	11.76	13.45	11.76
TIPS 2.000	15 Jul 2014	0.03	–0.46	0.03	–0.46	0.03	–0.46
TIPS 1.625	15 Jan 2015	–1.45	–1.89	–1.42	–1.89	–1.47	–1.94
TIPS 1.875	15 Jul 2015	–0.66	–1.44	–0.14	–0.95	–0.63	–1.44
TIPS 2.000	15 Jan 2016	–2.56	–3.68	–2.46	–3.58	–2.52	–3.64
TIPS 2.500	15 Jul 2016	0.27	–1.19	0.83	–0.66	0.83	–0.66
TIPS 2.375	15 Jan 2017	2.01	0.79	2.80	1.58	2.50	1.29
TIPS 2.625	15 Jul 2017	0.55	–0.35	1.22	0.32	1.22	0.32
TIPS 1.625	15 Jan 2018	1.95	0.96	2.95	1.96	2.95	1.96
TIPS 1.375	15 Jul 2018	2.53	2.41	2.53	2.41	2.53	2.41
TIPS 2.125	15 Jan 2019	4.45	3.71	1.95	1.21	1.95	1.21
TIPS 2.375	15 Jan 2025	0.55	–0.23	0.83	0.05	3.44	2.64
TIPS 2.000	15 Jan 2026	3.06	2.29	–1.02	–1.78	3.12	2.36
TIPS 2.375	15 Jan 2027	6.58	5.68	6.87	5.97	6.75	5.85
TIPS 1.750	15 Jan 2028	1.74	0.14	2.35	0.75	2.29	0.69
TIPS 3.625	15 Apr 2028	–6.92	–7.88	–7.79	–8.75	–7.83	–8.97
TIPS 2.500	15 Jan 2029	–4.60	–4.60	–9.38	–9.38	–9.38	–9.38
TIPS 3.875	15 Apr 2029	–3.63	–4.50	–4.55	–5.36	–4.54	–5.35
TIPS 3.375	15 Apr 2032	–2.41	–3.29	–2.80	–3.68	–4.20	–5.08

Notes: See notes to Table 3. This table presents the excess returns (ER) of the long–short trading strategy on the basis of the Y_1 forecasting horizon and different holding periods.

Table 5 presents the results for the breakeven trading strategy with Survey forecasts for Q_0.

Breakeven trades with a holding period of three months provide positive excess returns across all series of TIPS. The exceptions are the most recent TIPS issues with less than six months of data history. Positive excess returns are also shown for holding periods of 6 and 12 months. Although the 3-month holding period is the most profitable strategy across the TIPS issues, the 12-month holding period leads to the highest excess returns. These findings are in line with those of Bardong and Lehnert (2008). Excess returns with Q_1 forecasts are similar (results not reported). Excess returns increase in magnitude, and we also observe more TIPS issues with negative excess returns.

Therefore, the breakeven trading strategy with signals based on Survey data is profitable over the entire maturity spectrum of the TIPS market. Implementing this strategy leads to excess returns of approximately 3.5 percent a year (with transaction costs). Empirical tests also confirm the profitability of the breakeven strategy when it is used with one-year-ahead inflation forecasts.

The positive outperformance of the breakeven strategy with our inflation-forecasting model is not as pronounced as the outperformance generated by using inflation forecasts from the Survey. Although breakeven trades on Y_1 forecasts resulted in negative returns for five series of TIPS, negative returns occurred for only two series of TIPS when the Q_0 forecasting horizon was used. Further, the average

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Table 5. Performance of Breakeven Strategy with Q_0 Forecasting Horizon

		ER with 3-Mo. HP		ER with 6-Mo. HP		ER with 12-Mo. HP	
Name	Maturity Date	Without TC (%)	With TC (%)	Without TC (%)	With TC (%)	Without TC (%)	With TC (%)
TIPS 3.375	15 Jan 2007	4.33	3.91	4.32	3.89	4.34	3.91
TIPS 3.625	15 Jan 2008	3.56	3.04	3.60	3.09	3.64	3.10
TIPS 3.875	15 Jan 2009	5.48	4.79	5.10	4.39	5.30	4.59
TIPS 4.250	15 Jan 2010	4.39	3.88	4.35	3.84	4.39	3.88
TIPS 0.875	15 Apr 2010	2.88	2.29	2.98	2.35	2.97	2.34
TIPS 3.500	15 Jan 2011	4.26	3.75	4.19	3.67	5.59	5.07
TIPS 2.375	15 Apr 2011	0.91	0.49	1.65	1.17	3.25	2.77
TIPS 3.375	15 Jan 2012	4.85	4.25	4.81	4.20	4.80	4.18
TIPS 2.000	15 Apr 2012	0.00	0.00	0.00	0.00	0.00	0.00
TIPS 3.000	15 Jul 2012	4.94	4.24	4.86	4.15	5.01	4.30
TIPS 0.625	15 Apr 2013	2.05	1.13	1.82	0.90	0.52	–0.40
TIPS 1.875	15 Jul 2013	4.28	3.75	4.22	3.68	4.29	3.75
TIPS 2.000	15 Jan 2014	5.44	5.03	5.36	4.98	5.25	4.86
TIPS 1.250	15 Apr 2014	12.44	10.17	12.44	10.17	12.44	10.17
TIPS 2.000	15 Jul 2014	5.67	5.07	5.68	5.06	5.80	5.15
TIPS 1.625	15 Jan 2015	5.03	4.17	5.12	4.23	5.17	4.23
TIPS 1.875	15 Jul 2015	3.98	2.97	2.97	1.94	4.03	2.98
TIPS 2.000	15 Jan 2016	4.04	2.89	4.14	2.96	4.21	2.97
TIPS 2.500	15 Jul 2016	5.55	4.43	5.70	4.58	5.70	4.58
TIPS 2.375	15 Jan 2017	5.35	4.14	5.54	4.32	5.39	4.17
TIPS 2.625	15 Jul 2017	6.63	5.61	6.87	5.86	6.87	5.86
TIPS 1.625	15 Jan 2018	4.69	3.78	4.33	3.41	4.21	3.30
TIPS 1.375	15 Jul 2018	0.00	0.00	0.00	0.00	0.00	0.00
TIPS 2.125	15 Jan 2019	0.45	–0.29	–0.53	–1.27	–0.53	–1.27
TIPS 2.375	15 Jan 2025	11.80	11.44	12.08	11.72	12.21	11.85
TIPS 2.000	15 Jan 2026	2.73	2.15	3.18	2.61	3.07	2.49
TIPS 2.375	15 Jan 2027	1.30	0.98	1.45	1.14	1.25	0.93
TIPS 1.750	15 Jan 2028	8.44	7.53	8.45	7.54	8.15	7.24
TIPS 3.625	15 Apr 2028	9.00	8.68	9.00	8.67	9.72	9.39
TIPS 2.500	15 Jan 2029	–15.20	–16.44	–13.62	–14.85	–13.62	–14.85
TIPS 3.875	15 Apr 2029	5.95	5.66	5.75	5.46	5.82	5.53
TIPS 3.375	15 Apr 2032	5.87	5.54	5.58	5.24	6.08	5.75

Notes: See notes to Table 3. This table presents the excess returns (ER) of the breakeven trading strategy on the basis of the Q_0 forecasting horizon and different holding periods.

returns for Y_1, Y_2, and Y_3 forecasts were approximately 1.5 percent, which is less than the outperformance when the Survey was used.

Thus, the breakeven strategy performs better than the long–short strategy under either the Survey or our inflation-forecasting model. With the breakeven trading strategy, excess returns are higher when inflation forecasts from the Survey are used. Q_0 and Q_1 breakeven trades outperform Y_1 trades for short- and medium-term TIPS but underperform for long-term TIPS. The best performing strategy uses Q_0 forecasts with a 12-month holding period, Q_1 forecasts with a 6-month holding period, and Y_1 forecasts with a 12-month holding period.

We also examined TIPS performance during 2008, a period of increased macroeconomic uncertainty and market volatility. Not only was 2008 the year with the highest inflation rate since TIPS were issued (see Figure 1), but it was also characterized by financial turmoil unprecedented in the TIPS market. We studied the performance of the breakeven strategy during this period. Since the beginning of the TIPS market, the highest inflation rate was reached in the last quarter of 2007 and in the first three quarters of 2008. During this one-year period, breakeven trades resulted in positive excess returns across all series of TIPS. The positive returns were highest in the third quarter of 2008, when the inflation rate in July and August was above 5 percent. The breakeven strategy produced positive returns in the third quarter of 2008 for all 32 series of TIPS. The main reason for these results is that TIPS

became more attractive as the uncertainty regarding future changes in inflation increased and inflation risk premiums became more important. In these unstable macroeconomic conditions, however, TIPS market participants' predictions of the future inflation rate turned out to be rather poor. Our admittedly simple inflation forecasts seem to provide more precise inflation predictions. Hence, the use of systematic inflation forecasts seems to be particularly beneficial to TIPS investors during periods of macroeconomic uncertainty.

Furthermore, comparing the performance of the TIPS market with the performance of the nominal Treasury bond market is also an interesting case study because it unveils some important drivers of TIPS prices. Both nominal and real Treasury bonds are virtually default free and should thus provide investors with the most secure returns during financial crises. Surprisingly, we observed significant negative returns for long BEI trades during this period, especially with short-maturity TIPS. In addition, BEI was negative during that time because the nominal Treasury rates (e.g., the yields on STRIPS) were lower than the yields on the maturity-matched TIPS. Negative BEI implies that investors expect deflation until the respective TIPS mature. This particular expectation is unlikely to be a normal macroeconomic condition and, therefore, is not a realistic measure of inflation expectations. Given that TIPS provide for deflation protection, observing negative BEI is even more puzzling. Negative BEI seems to be the result of a dramatic increase in the liquidity premium, one of the major drivers of TIPS prices mentioned earlier. Apparently, TIPS are not considered equivalent to or as liquid as their matching nominal bonds because the financial crisis resulted in a flight to liquidity among both individual and institutional investors.

In contrast to the TIPS-implied inflation rates, we found a positive expected one-year-ahead inflation rate from our inflation forecasts. With respect to the trading strategies, this finding suggests that the forecasted inflation rate was constantly higher than BEI during the last quarter of 2008 and the first two quarters of 2009. Thus, the breakeven strategy between the last quarter of 2008 and the second quarter of 2009 resulted in the same positions as the comparable long-only strategy. So, no TIPS series on the short and intermediate sections of the TIPS maturity spectrum earned excess returns over the respective return benchmarks during these three quarters. Holdings began to differ from the benchmark strategy only for the TIPS series with longer maturities from May 2009.

Another important observation is that the breakeven strategy shows negative excess returns for medium- to long-term TIPS from the beginning of 2009 (see Table 5 and **Table 6**). This finding suggests that the strategy did not perform well in the economic downturn because inflation expectations were adjusted downward.

In sum, during the 2008 financial crisis, the breakeven strategy kept a long BEI position because the increased liquidity premium pushed BEI far below a realistic inflation forecast. This finding implies that TIPS are not considered equivalent to nominal government bonds. Investors seem to ignore their security and inflation protection and apparently prefer the highest level of liquidity. The conclusion reached by Sack and Elsasser (2004) is, therefore, still valid: TIPS are unlikely to be as liquid as nominal Treasuries because of their different roles in the financial markets. The liquidity premium required by investors for holding TIPS also appears to vary over time, depending on macroeconomic conditions and financial market stability.

Conclusion

The continuous existence of positive excess returns provides evidence of persistent TIPS market inefficiencies, which can be exploited by active trading in the inflation-indexed bond markets. We found that the Survey of Professional Forecasters provides valuable information that can be exploited through breakeven trades. Furthermore, long-term forecasting models seem to provide additional valuable information. Generating positive excess returns from trading TIPS on a long–short basis by using our simple inflation forecasts appears to be more challenging, which confirms previous empirical findings (see, e.g., Bardong and Lehnert 2004a, 2004b, 2008).

Because the Survey is publicly available, realized excess returns based on Survey information are an argument against TIPS market efficiency. Breakeven trades that use Survey information show positive excess returns of up to 3.5 percent a year (with transaction costs). Breakeven trades that use inflation forecasts appear to be profitable in the short, intermediate, and long sections of the TIPS maturity spectrum.

A more detailed analysis of trading strategy–related excess returns suggests that the breakeven strategy performs well during periods of high macroeconomic volatility. This finding can be explained by an enhanced attractiveness of TIPS as uncertainty around future inflation increases. This finding also suggests that, in general, TIPS market participants' predictions of future inflation rates during periods of macroeconomic uncertainty are rather poor. Using a systematic approach to forecast inflation is, therefore, especially useful during times of economic uncertainty.

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Table 6. Performance of Breakeven Strategy with Y_1 Forecasting Horizon

		ER with 3-Mo. HP		ER with 6-Mo. HP		ER with 12-Mo. HP	
Name	Maturity Date	Without TC (%)	With TC (%)	Without TC (%)	With TC (%)	Without TC (%)	With TC (%)
TIPS 3.375	15 Jan 2007	4.48	4.06	4.47	4.04	5.46	5.03
TIPS 3.625	15 Jan 2008	2.14	1.70	1.87	1.43	2.22	1.78
TIPS 3.875	15 Jan 2009	2.46	2.07	2.10	1.71	3.07	2.68
TIPS 4.250	15 Jan 2010	2.13	1.70	1.50	1.05	3.14	2.70
TIPS 0.875	15 Apr 2010	0.28	0.05	0.69	0.46	1.26	1.02
TIPS 3.500	15 Jan 2011	3.64	2.91	3.11	2.37	5.15	4.41
TIPS 2.375	15 Apr 2011	0.00	0.00	0.00	0.00	0.00	0.00
TIPS 3.375	15 Jan 2012	3.77	3.12	3.66	3.04	4.79	4.16
TIPS 2.000	15 Apr 2012	–0.38	–0.49	–0.38	–0.49	–0.38	–0.49
TIPS 3.000	15 Jul 2012	2.41	1.84	1.18	0.61	1.18	0.61
TIPS 0.625	15 Apr 2013	3.54	2.90	3.31	2.67	1.15	0.51
TIPS 1.875	15 Jul 2013	2.76	2.74	4.21	3.67	5.67	5.13
TIPS 2.000	15 Jan 2014	2.96	2.31	2.82	2.17	2.74	3.38
TIPS 1.250	15 Apr 2014	–1.99	–3.68	–1.99	–3.68	–1.99	–3.68
TIPS 2.000	15 Jul 2014	2.85	2.36	2.85	2.36	2.85	2.36
TIPS 1.625	15 Jan 2015	2.24	1.79	2.35	1.88	2.33	1.86
TIPS 1.875	15 Jul 2015	3.67	2.89	3.73	2.92	3.37	2.56
TIPS 2.000	15 Jan 2016	4.21	3.09	5.07	3.95	4.96	3.85
TIPS 2.500	15 Jul 2016	6.31	4.82	6.51	5.01	6.51	5.01
TIPS 2.375	15 Jan 2017	2.53	1.31	2.30	1.08	8.56	7.34
TIPS 2.625	15 Jul 2017	2.88	1.98	2.90	2.00	2.90	2.00
TIPS 1.625	15 Jan 2018	–0.63	–1.62	–0.99	–1.98	–1.11	–2.10
TIPS 1.375	15 Jul 2018	3.50	3.39	3.50	3.39	3.50	3.39
TIPS 2.125	15 Jan 2019	–22.41	–23.15	–23.39	–24.13	–23.39	–24.13
TIPS 2.375	15 Jan 2025	10.04	9.26	10.22	9.45	10.30	11.10
TIPS 2.000	15 Jan 2026	8.20	7.43	10.12	9.35	14.30	13.53
TIPS 2.375	15 Jan 2027	11.39	10.49	11.42	10.52	11.22	10.32
TIPS 1.750	15 Jan 2028	25.62	24.03	25.64	24.04	25.34	23.74
TIPS 3.625	15 Apr 2028	12.05	11.09	11.99	11.03	12.33	11.37
TIPS 2.500	15 Jan 2029	–70.49	–70.49	–68.90	–68.90	–68.90	–68.90
TIPS 3.875	15 Apr 2029	5.74	4.93	6.04	5.23	6.40	5.60
TIPS 3.375	15 Apr 2032	10.98	10.10	11.59	10.71	12.22	11.35

Notes: See notes to Table 3. This table presents the excess returns (ER) of the breakeven trading strategy on the basis of the Y_1 forecasting horizon and different holding periods.

Our results could be useful for investors that prefer predictable real cash flows, such as pension funds and people saving for retirement. Theoretically, index-linked government bonds are among the most straightforward inflation hedges available today. Owing to market inefficiencies and the possibility of changes in real yields over time, however, inflation-indexed bonds may well introduce, rather than eliminate, risks to investors' portfolios.

Our results suggest that the TIPS market has yet to fulfill investors' expectations of being a low-risk, efficient, and liquid financial instrument. We consistently found positive excess returns, even after accounting for trading costs, for one of the investment strategies we used, which implies that the TIPS market is not yet fully efficient. Also, as our results for 2008 show, additional risk factors, such as liquidity and macroeconomic uncertainty, can sometimes be important.

With respect to the financial crisis between the last quarter of 2008 and the second quarter of 2009, our observations further point to inefficiencies in the TIPS market. The most striking evidence from our results is that BEI turns negative. This finding implies that TIPS are not considered equivalent to government bonds, which leads to the requirement of a substantial liquidity premium for holding TIPS. Moreover, the liquidity premium appears to change significantly over time and to vary with such factors as perceived financial market stability.

This article qualifies for 1 CE credit.

Notes

1. U.S. Department of the Treasury, Bureau of the Public Debt (www.publicdebt.treas.gov).
2. The semi-strong form of market efficiency implies that security prices reflect all publicly available information (Fama 1970).
3. Early studies of index-linked bonds matched nominal and real bonds by duration. But real cash flows of index-linked bonds are fixed, whereas those of nominal bonds decline over time (Sack 2000). Nominal bonds thus have a shorter duration than index-linked bonds. In addition, although index-linked bonds face real interest rate risk, nominal bonds are exposed to nominal interest rate risk (Spiegel 1998). Bootle (1991) argued that these two forms of interest rate risk differ because of changes in the relative size of inflation risk premiums and real rates over time. The market convention of maturity matching TIPS with nominal bonds arguably leads to a duration mismatch because the back-loaded nominal cash flow of TIPS results in their having a longer duration than maturity-matched nominal bonds. Determining the nominal duration of TIPS—or inflation-indexed bonds in general—is difficult (see, e.g., Dektar 2005), although attempts have been made to provide an effective hedge to TIPS' nominal interest rate exposure (see, e.g., Rudolph-Shabinsky and Trainer 1999; Laatsch and Klein 2005). Because current market convention is to use maturity matching, we followed market practice to ensure that our results would be relevant to market practitioners. Using STRIPS (Separate Trading of Registered Interest and Principal Securities) for maturity matching may help alleviate the duration mismatch because zero-coupon STRIPS have a longer duration than coupon Treasuries with the same maturity.
4. When nominal Treasuries or TIPS are stripped, each interest payment and the principal payment become a separate zero-coupon security.
5. Matching TIPS with zero-coupon bonds may create a liquidity mismatch, which could be driving our results. Thus, we repeated our analysis by using on-the-run nominal and inflation-protected bonds to focus on the most liquid instruments of each type of bond. To that end, we constructed on-the-run TIPS for both the 5- and 10-year maturity buckets. We did not create an on-the-run TIPS series for the 20-year maturity bucket because nominal 20-year Treasury coupon bonds were issued only irregularly during the period covered by our study. The resulting on-the-run 10-year TIPS series comprises data from 19 individual issues from January 1997 to June 2009. The on-the-run 5-year TIPS series is based on data from five individual bonds issued between October 2004 and June 2009. We maturity matched these on-the-run TIPS with on-the-run Treasury coupon bonds. We matched the on-the-run 10-year Treasury series, typically issued quarterly (in February, May, August, and November), with the on-the-run 10-year TIPS series. Likewise, we matched the on-the-run 5-year Treasury series with the on-the-run 5-year TIPS series. Our results (available on request) are largely consistent with our findings for bond-level maturity matching. In particular, both the long–short strategy and the breakeven strategy typically generate positive excess returns. We also found that the breakeven strategy tends to outperform the long–short strategy. This finding provides some evidence that our results are not driven entirely by liquidity premiums. That we found some instances of very small to negative average excess returns, however, could be interpreted as an indication that a substantial portion of our excess returns is related to liquidity premiums.
6. Because this feature increases return volatility, some researchers exclude from their empirical analysis TIPS with a maturity of less than two years (see, e.g., Gürkaynak, Sack, and Wright 2007). But because we looked at the profitability of trading TIPS over their entire lifetimes, we considered the inclusion of these bonds important.
7. We obtained the Survey data from the Federal Reserve Bank of Philadelphia website (www.philadelphiafed.org/research-and-data/real-time-center/survey-of-professional-forecasters).
8. Assuming that inflation is realized homogeneously throughout the month, monthly inflation index levels are converted into weekly numbers by linearly interpolating between the monthly observations. If a week falls between two months, its inflation index level is calculated as the weighted average of the weekly inflation index levels of the two months. The weekly inflation rates are calculated as the log difference between two index levels 52 weeks apart. Finally, the daily inflation rate is defined as one-fifth of the weekly inflation numbers (see also Bardong and Lehnert 2004a).

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